🧠 Multi-Agent Systems Quiz

📌 Section A: MCQs (Conceptual Understanding)

1. What is the primary difference between a single-agent system and a multi-agent system?
   A. Single-agent systems use AI, multi-agent systems do not
   B. Multi-agent systems involve multiple interacting agents
   C. Single-agent systems are always faster
   D. Multi-agent systems cannot make decisions independently

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2. In a multi-agent system, agents are typically:
A. Completely dependent on each other
B. Independent but capable of interaction
C. Controlled by a central authority only
D. Static and non-adaptive

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3. Which of the following is an example of a multi-agent system?
A. Calculator app
B. Self-driving car network communicating with other cars
C. Text editor
D. Single chatbot

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4. What is a key advantage of multi-agent systems?
A. Simpler design
B. No communication required
C. Scalability and parallelism
D. Limited decision-making

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5. Which architecture uses only one decision-making entity?
A. Distributed system

B. Multi-agent system
C. Single-agent system
D. Hybrid system

📌 Section B: Short Answer Questions

6. Define a multi-agent system in your own words
Ans. A **multi-agent system** is a system where **multiple intelligent agents work together**, each performing a specific task, to solve a complex problem or achieve a common goal.

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7. List two advantages of multi-agent systems over single-agent systems
Ans. 1. Parallel execution
Multiple agents can work at the same time, making the system faster.
2. Specialization
Each agent focuses on a specific task, leading to better quality results.

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8. What is agent communication, and why is it important?
Ans. **Agent communication** is the process by which agents **share information and messages with each other**.
It is important because:

- It helps agents **coordinate tasks**
- Ensures **correct decision-making**
- Allows **collaboration between agents**

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9. Name two types of agent communication patterns
Ans. **1. Sequential (Pipeline)**
Agents work one after another (e.g., Search → Analyst → Writer)
2. Broadcast (Parallel)
One agent sends information to all agents at the same time

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10. What is task delegation in multi-agent systems?
Ans. **Task delegation** is the process where one agent (usually a coordinator) **assigns tasks to other agents based on their roles or capabilities**.

📌 Section C: Analytical / Application-Based Questions

11. Compare single-agent and multi-agent systems based on:

- Decision-making
- Scalability
- Complexity

Ans 11. Single-agent vs Multi-agent

- **Decision-making:**
  Single → one agent
  Multi → multiple agents

- **Scalability:**
  Single → low
  Multi → high

- **Complexity:**
  Single → simple
  Multi → complex

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12. Explain the difference between **centralized** and **decentralized** multi-agent architectures.

- Ans 12. **Centralized:** One controller manages all agents

- **Decentralized:** Agents work independently without a central controller

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13. A group of delivery robots must coordinate to deliver packages efficiently.

- What type of system is this?
- Why is communication important here?

- Ans 13. **Type:** Multi-agent system

- **Importance of communication:** Coordination, avoid conflicts, efficient delivery

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14. If one agent assigns tasks to others based on their capabilities, what strategy is being used?
A. Broadcasting
B. Task delegation
C. Isolation
D. Random assignment

Ans 14. **B. Task delegation**

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15. Explain how **peer-to-peer communication** works in multi-agent systems.

Ans 15. Agents communicate **directly with each other** without a central controller

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📌 Section D: Scenario-Based Questions

16. In a smart traffic system, traffic lights (agents) communicate with each other to reduce congestion.

- Identify the communication pattern used
- What could happen if communication fails?

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17. Consider a team of AI agents solving a complex problem:

- One agent plans
- Others execute tasks

What architecture or strategy is this? Explain.

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18. A multi-agent system uses broadcasting to share information.

- What are the advantages and disadvantages of this approach?

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19. In a distributed AI system, each agent has partial knowledge of the environment.

- How can agents collaborate effectively?

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20. Suppose a system fails when one central controller crashes.

- What type of architecture is this?
- Suggest a better alternative.

ANSWERS 16-20:

16. Smart Traffic System

- **Communication pattern:** Peer-to-peer (or decentralized)
- **If communication fails:** Traffic jams, poor coordination, increased congestion

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17. AI Agents Team

Architecture/Strategy: Centralized (or hierarchical)
Explanation: One agent plans (controller), others execute assigned tasks

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18. Broadcasting

- **Advantages:** Fast information sharing, all agents updated simultaneously
- **Disadvantages:** High communication cost, possible information overload

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19. Partial Knowledge Collaboration

Agents collaborate by:

- Sharing information
- Communicating frequently
- Coordinating decisions

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20. System Failure

- **Type:** Centralized architecture
- **Better alternative:** Decentralized architecture (no single point of failure)

Section E: Higher-Order Thinking

21. Design a simple multi-agent system for **online food delivery**.

- Define roles of at least 3 agents

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22. Why is task delegation important in large-scale systems like cloud computing?

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23. Compare **cooperative vs competitive agents** with examples.

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24. How does communication overhead impact system performance?

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25. Suggest improvements for a poorly coordinated multi-agent system.

ANSWERS 21-25:

21. Multi-Agent System for Food Delivery

Agents & Roles:

- **Order Agent:** Takes customer order
- **Restaurant Agent:** Prepares food
- **Delivery Agent:** Delivers food to customer

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22. Importance of Task Delegation

It helps:

- Distribute workload
- Improve efficiency
- Use resources effectively

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23. Cooperative vs Competitive Agents

- **Cooperative:** Work together for common goal
  Example: Delivery agents coordinating routes
- **Competitive:** Compete against each other
  Example: Online bidding systems

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24. Communication Overhead Impact

High communication:

- Slows down system
- Uses more resources
- Reduces efficiency

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25. Improvements in Poorly Coordinated System

Suggestions:

- Improve communication
- Define clear roles
- Use better coordination strategies
- Add centralized or hybrid control